

Context Length

- We've seen that Markov models (for language modeling, prediction in many cases etc.) are effective, but do not capture dependencies beyond a certain length
- By the Markov assumption:
 - It's enough to know the last n tokens you've encountered to know what's next
 - Alternative view: the probability of a sequence is the product of the probabilities of its *n-grams*

The brown cat that disturbed my sleep yesterday _____ (is/are)

The brown cats that disturbed my sleep yesterday _____ (is/are)

Sentiment Analysis as Sequence Labeling

- Consider the example:
 - *How can you not see this movie?*
 - *You should not see this movie*
- How well will a bi-gram model work?
 - Similar unigrams and bigrams → similar prediction
- The problem with Markov Models: need to maintain a **state** to capture distant influences
 - The size of the space increases exponentially with the order of the model

Recall: Markov Models are Finite State Automata (FSAs) with transition probabilities

